Heart Failure

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Microbiome Signatures, Microbiome Medicine

Abstract. The gut microbiome significantly influences heart failure progression, contributing to inflammation and other complications.

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Overview

Heart failure (HF) is a complex syndrome affecting millions worldwide, characterized by the heart's inability to fill or eject blood properly. Emerging research highlights the role of gut microbiota in HF, suggesting that changes in gut metabolites are closely related to HF progression. The gut hypothesis of HF posits that reduced cardiac output and systemic congestion lead to intestinal ischemia and barrier dysfunction, resulting in bacterial translocation and inflammation. We explore the microbiome signature and gut microbiota's involvement in HF, mechanisms mediated by gut metabolites, as well as potential interventions, including dietary changes, probiotics, fecal microbiota transplantation (FMT), and antibiotics.

Gut Hypothesis of Heart Failure

The gut hypothesis of HF, first introduced in the late 1990s, proposed that circulatory congestion and low cardiac output reduce intestinal perfusion, causing ischemia and intestinal barrier damage, leading to bacterial translocation, endotoxemia, and systemic inflammation. [1][2] Several studies have further supported the idea that metabolites like trimethylamine-n-oxide (TMAO) and SCFAs, produced by the gut microbiota, significantly impact HF progression. [3]

Primer

Research indicates that heart failure (HF) patients exhibit distinct gut microbiota profiles compared to healthy individuals. Key findings reveal an increased prevalence of pro-inflammatory bacteria such as *Bacteroides/Prevotella, Campylobacter*, *Shigella,Salmonella*, Enterococcus, and *Clostridium difficile* in HF patients. Conversely, there is a reduction in families like *Coriobacteriaceae*, *Erysipelotrichaceae*, and *Ruminococcaceae*, as well as anti-inflammatory genera such as *Blautia* and *Collinsella*. [4] Additionally, HF patients show reduced gut microbiota diversity, and an increased Firmicutes/ Bacteroidetes (F/B) ratio, which contributes to systemic conditions like persistent T-cell activation, and heightened susceptibility to hospitalization with a *Clostridium difficile* infection. [5][6]

Associated Conditions

Hospitalized HF patients are more frequently affected by *Clostridium difficile* infection, which is associated with in-hospital mortality. [7]

Metabolomic Signature of Heart Failure

Elevated metabolites contributing to heart failure (HF) have been identified, highlighting the influence of gut-derived compounds. Key metabolites include trimethylamine N-oxide (TMAO), which is produced from dietary nutrients by gut microbiota and is associated with cardiac fibrosis, hypertrophy, inflammation, and endothelial dysfunction. [8] Short-chain fatty acids (SCFAs), such as acetate, propionate, and butyrate, are derived from dietary fibers and provide energy to failing hearts, reduce inflammation, and prevent cardiac fibrosis and hypertrophy. [9]

Interventions

The gut hypothesis of HF underscores the potential of targeting gut microbiota for HF treatment. Changes in gut microbiota composition and metabolites like TMAO and SCFAs are significant in HF pathophysiology. Interventions such as dietary changes, probiotics, and possibly fecal microbiota transplantation FMT are often suggested as promising avenues for HF management.

Intervention	Findings
Pharmacological	
Rifaximin	Rifaximin is commonly used to treat microbiota toxicity and translocation by exerting anti-inflammatory effects and promoting the growth of beneficial bacteria like bifidobacteria and lactobacillus. Despite these potential benefits, the effects of antibiotics on gut microbiota in heart failure (HF) have not been extensively studied. It is crucial to balance the potential benefits and risks of antibiotic use.
Captopril	Captopril effectively modulates neurohormonal pathways, improves renal function, reduces cardiac workload, and corrects electrolyte imbalances, contributing to improved clinical outcomes. [10] While captopril's mechanism of action traditionally involves inhibiting the conversion of angiotensin I to angiotensin II, a potent vasoconstrictor, studies suggest that captopril modulates the gut microbiota, leading to a reduction in systemic inflammation and improvement in gut barrier function, both of which are implicated in the pathogenesis of HF. [11]
Drug Repurposing	
Metformin	Originally indicated to treat type 2 diabetes mellitus, metformin has been explored for its cardioprotective effects, primarily due to its ability to improve endothelial function, reduce oxidative stress, and exert anti-inflammatory effects. It may also improve myocardial energetics and reduce the risk of HF in diabetic and non-diabetic patients. Clinical trials are assessing metformin's efficacy in improving outcomes in patients with HF, particularly HF with preserved ejection fraction (HFpEF). [12]
SGLT2 inhibitors	SGLT2 inhibitors such as empagliflozin and dapagliflozin have shown significant benefits in reducing HF-related hospitalizations and cardiovascular mortality in both HF with reduced ejection fraction (HFrEF) and heart failure with preserved ejection fraction (HFpEF). [13] SGLT2 inhibitors are reported to influence gut microbiota composition, potentially by reducing glucose availability in the gut, which may inhibit pathogenic bacteria and support beneficial microbial populations. Preliminary studies in both animal models and humans suggest that these drugs can increase beneficial microbes and reduce pro-inflammatory bacteria, which might contribute to the observed reduction in inflammation and improved cardiovascular outcomes in heart failure patients treated with SGLT2 inhibitors. [14]
Spironolactone	Ongoing trials are investigating the effects of spironolactone in HFpEF and its potential to reduce fibrosis and improve diastolic function. As a mineralocorticoid receptor antagonist, spironolactone reduces aldosterone-induced sodium retention and cardiac remodeling. It is especially effective in reducing morbidity and mortality in HFrEF. [15]

Intervention	Findings
	Spironolactone can modulate the gut microbiota by reducing inflammation and improving the gut barrier function, potentially reducing the translocation of gut-derived endotoxins that contribute to systemic inflammation in HF. [16]
Colchicine	There is emerging evidence that colchicine may influence the gut microbiota by decreasing pro-inflammatory bacteria and promoting the growth of bacteria that produce SCFAs, which have anti-inflammatory effects. These changes could help mitigate the inflammatory response in HF. Further, colchicine significantly increases Firmicutes while reducing Bacteroidetes. Overall, colchicine intervention notably decreases the abundance of Bacteroidetes, Candida, and Clostridium, which are elevated in the microbiome signature of HF. [17]
Diet	
DASH Diet	A cohort study of 35,004 participants over 22 years indicated that the DASH diet reduces heart failure (HF) risk. In HF patients, the DASH diet improves walking test performance, arterial compliance, exercise capacity, and quality of life over a 3-month intervention. [18]
Non-Pharmacological	
Fecal microbiota transplantation (FMT)	Fecal microbiota transplantation (FMT) has primarily been used to treat recurrent *Clostridium difficile* infection, but holds promise for several other conditions hallmarked by the microbe. While the potential effects of FMT on HF are not well studied, FMT may hold promise as a supplementary treatment for HF given the consistent findings of *Clostridium difficile* in HF patients. [19]
Lactobacillus rhamnosus GR-1	Lactobacillus rhamnosus GR-1 can reduce hypertrophy and improve both systolic and diastolic functions of the left ventricle, indicating potential benefits for heart failure (HF) patients. [20]
Saccharomyces boulardii	A randomized, double-blind, placebo-controlled pilot trial targeting HF patients using *Saccharomyces boulardii* for 3 months showed that it could improve LVEF, shorten left atrial diameter, and lower total cholesterol and uric acid levels. [21] ***Saccharomyces boulardii*** can reduce filamentation, adhesion and biofilm formation of candida species, which are found elevated in HF patients. [22]
Probiotic yogurt	A triple-blind, controlled trial suggested that probiotic yogurt might help relieve inflammatory status in CHF patients by elevating sTWEAK levels, a cytokine involved in inflammation, tissue regeneration, and apoptosis. [23]
Supplements	

Intervention	Findings
Berberine	Oral intake of berberine for 4 months has been shown to decrease TMAO production in animal intestines and reduce TMA and TMAO levels in both the feces and plasma of patients, exerting effects similar to those of vitamins. [24]
Vitamin D+ B Vitamins	A study has suggested that combining B vitamins with vitamin D can alter choline metabolism, leading to a greater reduction in TMAO levels compared to the use of vitamin D alone. [25]
3,3-dimethyl-1-butanol (DMB)	DMB has been reported to improve cardiac function and reduce cardiac remodeling in heart failure (HF) mice induced by pressure overload. It achieves this by lowering plasma TMAO levels, which inhibits the TGF-β1/Smad3 and p65 NF-κB signaling pathways, thereby attenuating cardiac hypertrophy, fibrosis, and inflammation. [26]
High-Fiber Diet + Acetate Supplementation	High-Fiber Diet and acetate supplementation has been found to reduce the F/B ratio, while significantly increasing the abundance of the bacteria *Bacteroides acidifaciens*. This species has recently been shown to prevent obesity and the evolution of hypertension and heart failure in hypertensive mice.[27]
Microbiome Targeted Interventions (MBTIs)	
Carvacrol	Carvacrol (CR) shows potential as a candidate for microbiome-targeted interventions for heart failure due to its ability to positively modulate gut microbiota and reduce pathogen-induced dysbiosis. The study found that CR supplementation improved clinical outcomes in C. difficile infections, a pathogen also associated with the microbiome signature of HF. CR achieved this by increasing the abundance of beneficial bacteria like Firmicutes and reducing harmful bacteria like Proteobacteria, without significantly disrupting overall microbiome diversity. Given that gut dysbiosis and the presence of pathogens like C. difficile are linked to heart failure, CR's ability to restore healthy gut flora suggests it could be beneficial in managing gut microbiota imbalances associated with heart failure. [28] Hypertension-induced left ventricular hypertrophy is the most important risk factor for heart failure. This study finds that carvacrol was able to ameliorate cardiac hypertrophy in *in-vivo* and *in-vitro* models. [29]

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